Problem 6.1

If \$4,000 is invested at an annual rate of 6.0% compounded annually, what will be the final value of the investment after 10 years?

We use the compound interest formula:

$$FV = P(1+r)^n,$$

where

$$P = 4000, \qquad r = 0.06, \qquad n = 10.$$

Substituting into the formula:

$$FV = 4000(1+0.06)^{10}.$$

Therefore,

$$FV \approx 7163.39.$$

So, the final value of the investment is

$$\boxed{\$7163.39}.$$